

Comparison of FIR Filter Implementations: Direct, Genvar, and Optimized Methods (Pipelined vs Non-Pipelined)

1 Introduction

Finite Impulse Response (FIR) filters are essential components in digital signal processing used for noise reduction and signal shaping. In this work, FIR filters are implemented using three different methods: Direct implementation, Genvar-based implementation, and an Optimized implementation. Each method is further analyzed in both pipelined and non-pipelined architectures. The goal is to compare their resource utilization and performance.

2 Theory of Pipelining

Pipelining is a design technique used to improve the throughput of digital systems by dividing a computation into multiple stages separated by registers. Each stage performs a portion of the operation, allowing multiple data samples to be processed simultaneously.

In FIR filters, pipelining is introduced by placing registers between multiplication and addition stages. This reduces the critical path delay, enabling the system to operate at higher clock frequencies. As a result, the throughput increases significantly.

However, pipelining increases the number of registers, which leads to higher resource utilization. It also introduces latency, as the output is delayed by several clock cycles corresponding to the number of pipeline stages.

Despite the increased hardware usage, pipelining is advantageous because it allows stable high-speed operation, avoids timing violations, and improves overall system performance. Therefore, it is widely used in high-frequency digital systems.

3 FIR Filter Using IP Core

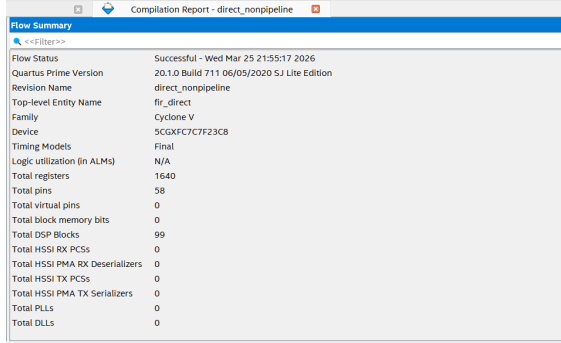
The FIR IP core provided by Intel is a highly optimized implementation of FIR filters. It uses internal pipelining, efficient multiplier usage, and DSP block mapping to achieve high performance.

The IP core operates using Avalon Streaming interfaces, which include valid signals for proper synchronization of data. It simplifies the design process by automatically generating optimized hardware based on user-defined parameters.

Compared to manual implementations, the FIR IP core provides better timing performance and efficient resource utilization, making it suitable for complex and high-speed applications.

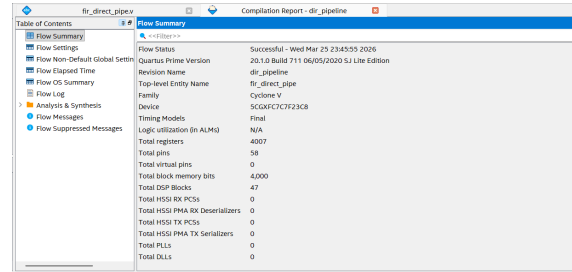
4 Resource Utilization Comparison

4.1 Direct Form Implementation



Flow Summary	
Flow Status	Successful - Wed Mar 25 21:55:17 2026
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition
Revision Name	direct_nonpipeline
Top-level Entity Name	fir_direct
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	1640
Total pins	58
Total virtual pins	0
Total block memory bits	0
Total DSP Blocks	99
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

(a) Non-Pipelined



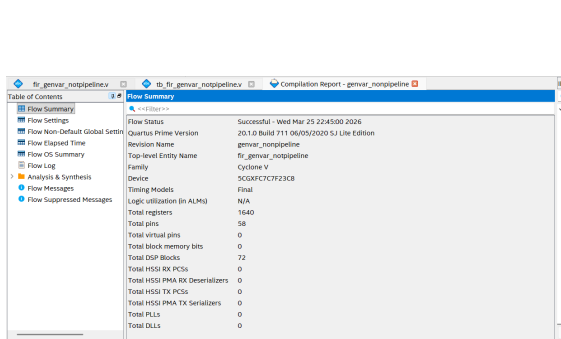
Flow Summary	
Flow Status	Successful - Wed Mar 25 23:45:55 2026
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition
Revision Name	dr_pipeline
Top-level Entity Name	fir_direct_pipe
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	4007
Total pins	58
Total virtual pins	0
Total block memory bits	4,000
Total DSP Blocks	47
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

(b) Pipelined

Figure 1: Resource Utilization for Direct Form FIR Filter

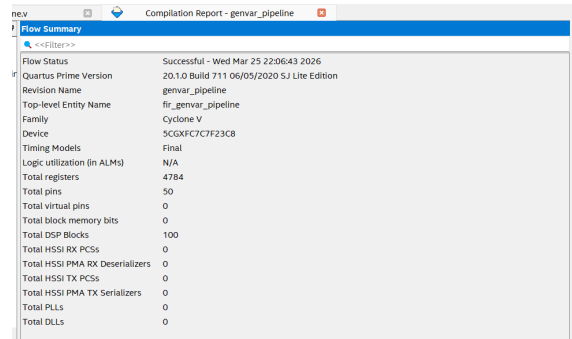
The direct form implementation shows that the pipelined design uses more registers compared to the non-pipelined design. However, the pipelined version improves timing performance by reducing the critical path.

4.2 Genvar-Based Implementation



Flow Summary	
Flow Status	Successful - Wed Mar 25 22:45:00 2026
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition
Revision Name	genvar_nonpipeline
Top-level Entity Name	fir_genvar_nonpipeline
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	1640
Total pins	58
Total virtual pins	0
Total block memory bits	0
Total DSP Blocks	72
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

(a) Non-Pipelined



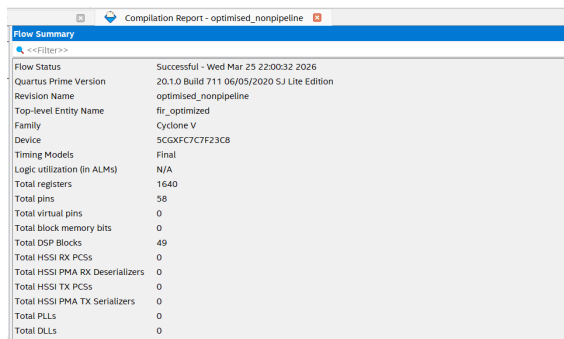
Flow Summary	
Flow Status	Successful - Wed Mar 25 22:06:43 2026
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition
Revision Name	genvar_pipeline
Top-level Entity Name	fir_genvar_pipeline
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	4784
Total pins	50
Total virtual pins	0
Total block memory bits	0
Total DSP Blocks	100
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

(b) Pipelined

Figure 2: Resource Utilization for Genvar-Based FIR Filter

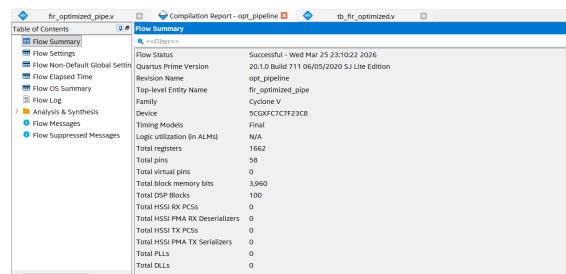
The genvar-based implementation provides a scalable design using generate constructs. Similar to the direct form, pipelining increases resource usage but improves performance.

4.3 Optimized Implementation



Flow Summary	
Flow Status	Successful - Wed Mar 25 22:00:32 2026
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition
Revision Name	optimised_nonpipeline
Top-level Entity Name	fir_optimized
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	1640
Total pins	58
Total virtual pins	0
Total block memory bits	0
Total DSP Blocks	49
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

(a) Non-Pipelined



Flow Summary	
Flow Status	Successful - Wed Mar 25 23:10:22 2026
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition
Revision Name	opt_pipeline
Top-level Entity Name	fir_optimized_pipe
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	1662
Total pins	58
Total virtual pins	0
Total block memory bits	3,960
Total DSP blocks	100
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

(b) Pipelined

Figure 3: Resource Utilization for Optimized FIR Filter

The optimized implementation reduces redundant computations and improves efficiency. Even though pipelining increases register count, it enables better utilization of hardware and higher operating speeds.

5 Discussion

From all three implementations, it is observed that pipelined designs consistently use more hardware resources than non-pipelined designs. This increase is mainly due to the additional registers introduced between stages.

However, pipelining significantly improves the maximum operating frequency by reducing the critical path delay. This makes pipelined designs more suitable for high-speed applications.

The optimized implementation shows better efficiency compared to direct and genvar methods, as it reduces redundant operations while maintaining performance.

6 Comparison of pipelined outputs with matlab

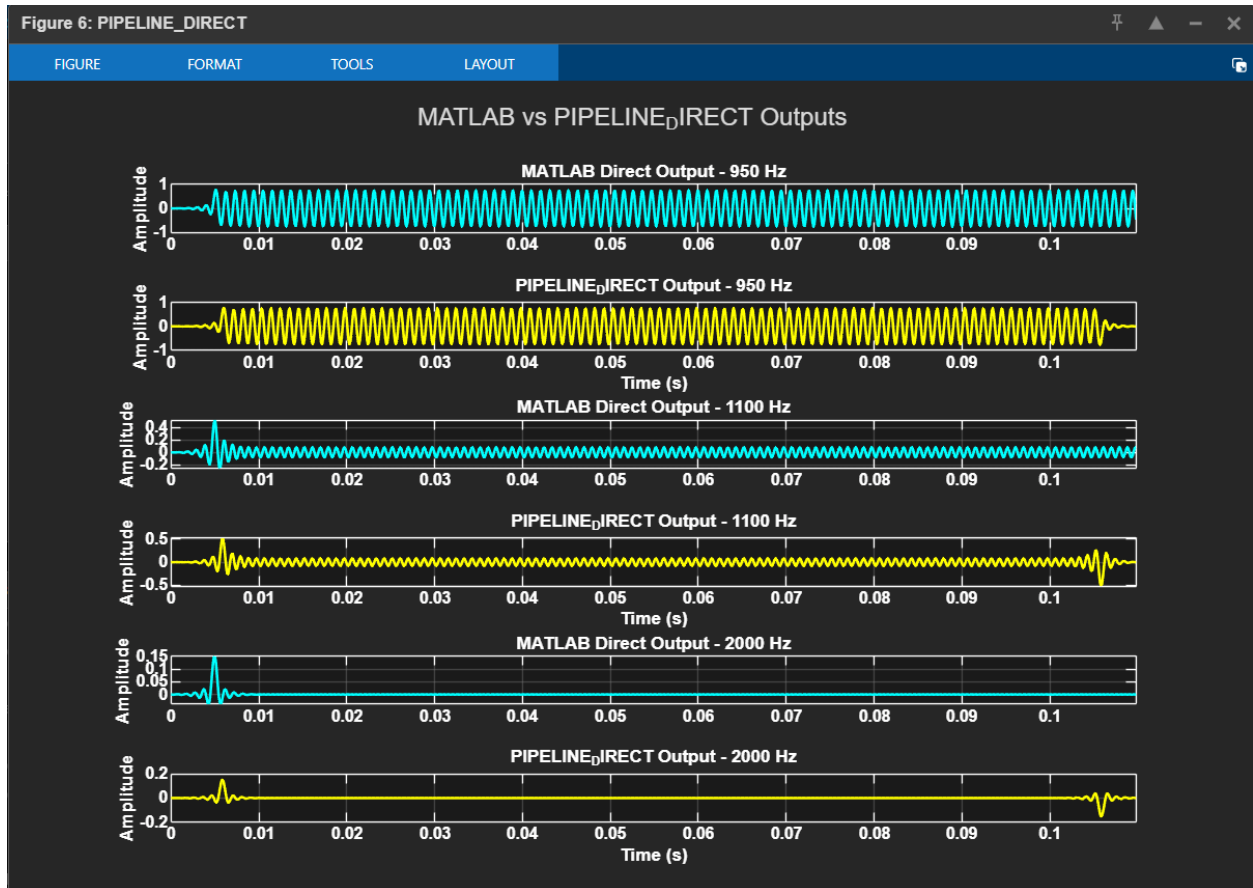


Figure 4

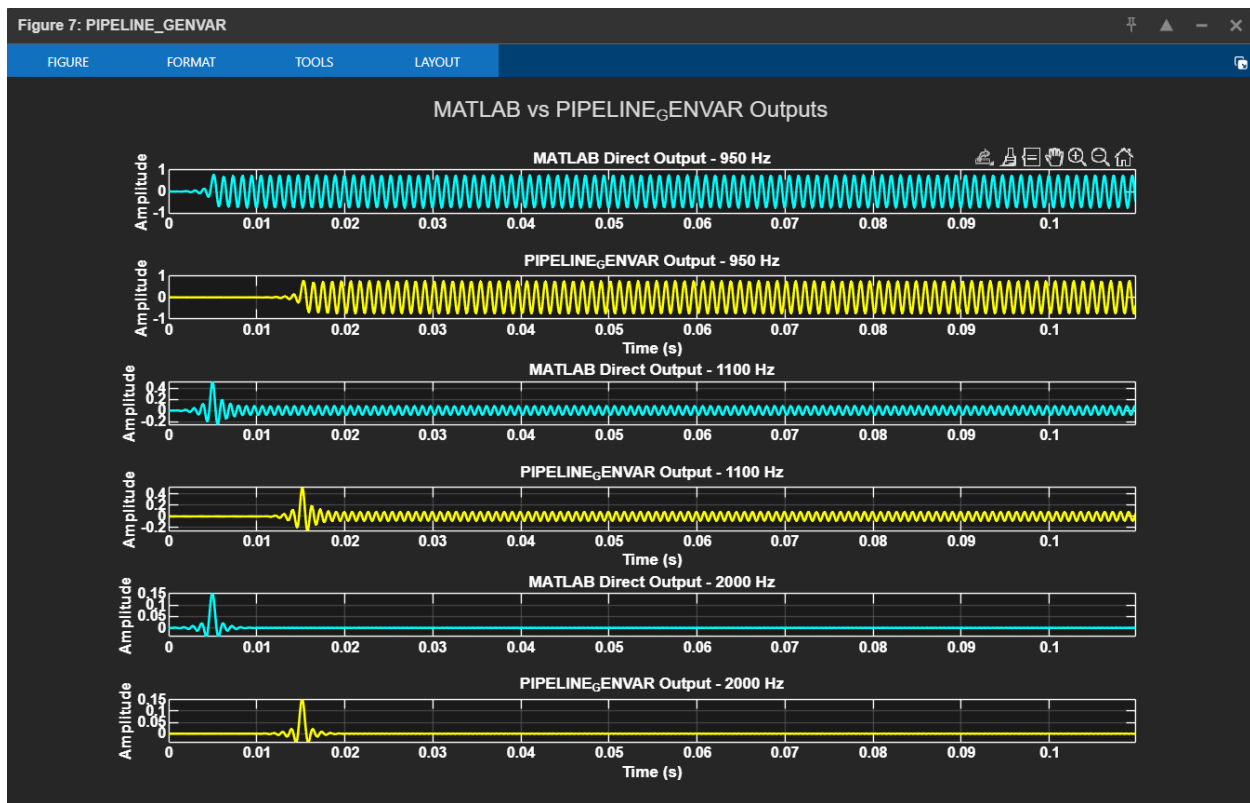


Figure 5

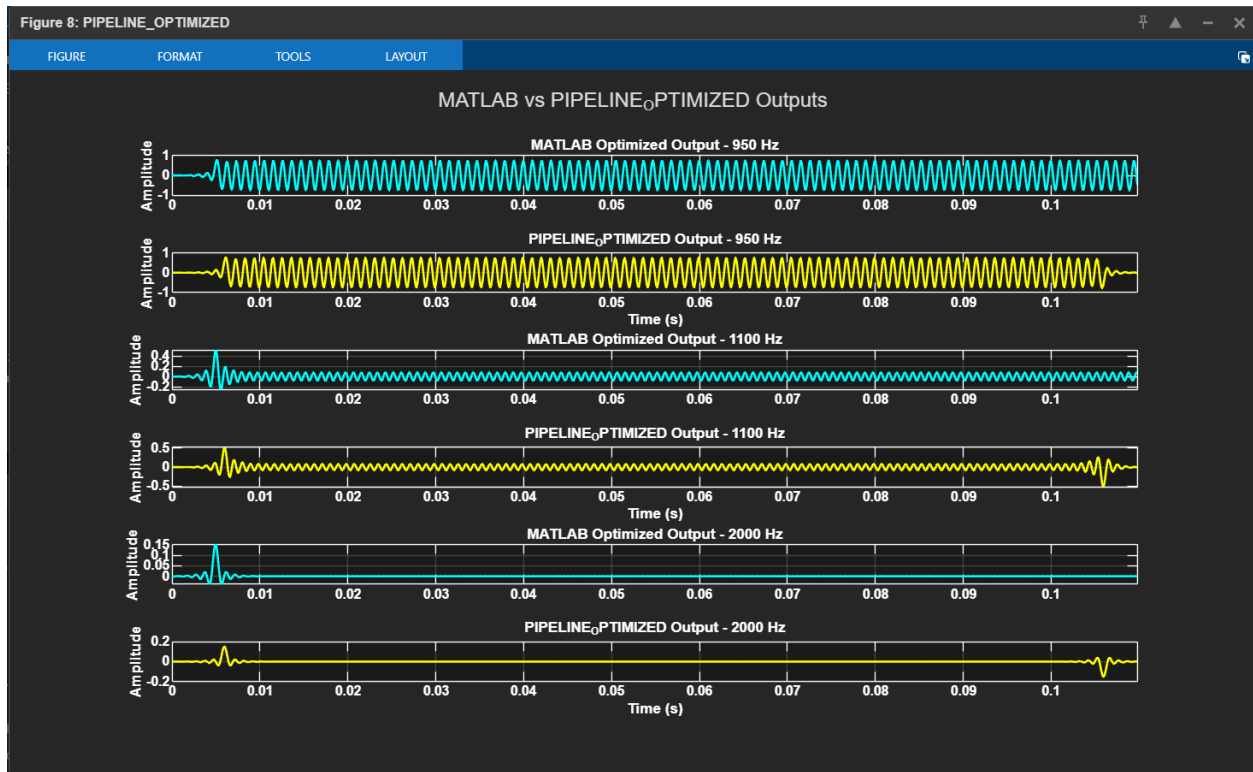


Figure 6

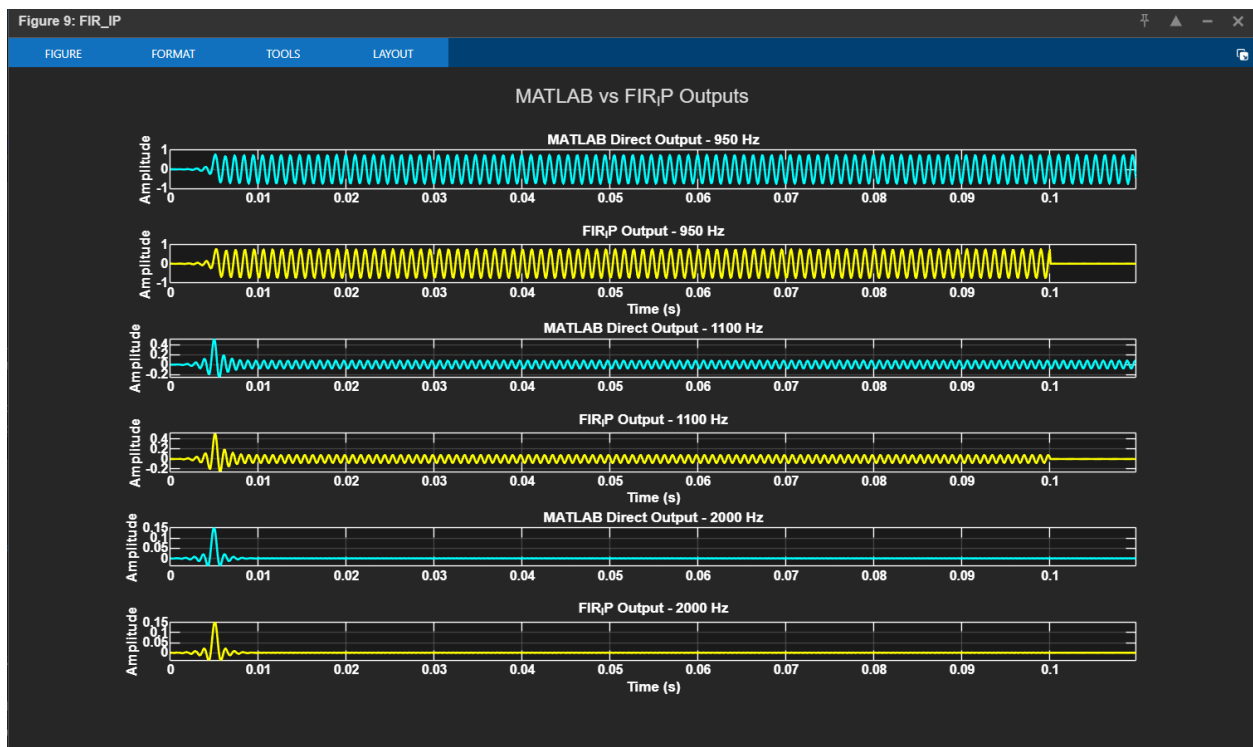
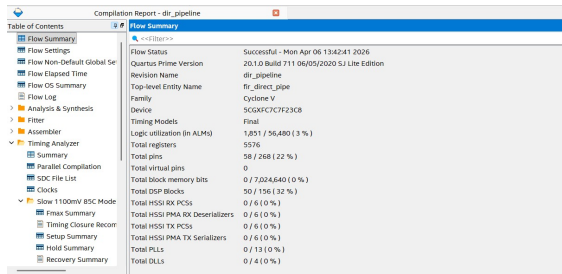


Figure 7

7 Direct Form Pipelined



Compilation Report - dir_pipeline

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Settings
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter
- Assembler
- Timing Analyzer
- Summary
- Parallel Compilation
- SDC File List
- Clocks
- Slow 1100mV 85C Model
- Fmax Summary
- Timing Closure Report
- Setup Summary
- Hold Summary
- Recovery Summary

Flow Summary

Flow Status: Successful - Mon Apr 06 13:42:41 2026

Quartus Prime Version: 20.1.0 Build 711 06/05/2020 5J Lite Edition

Revision Name: dir_pipeline

Top-level Entity Name: tr_direct_pipe

Family: Cyclone V

Device: 50GXFC7CF23C8

Timing Models: Final

Logic utilization (in ALMs): 1851 / 56480 (3 %)

Total registers: 5576

Total pins: 58 / 268 (22 %)

Total virtual pins: 0

Total block memory bits: 0 / 7,024,640 (0 %)

Total DSP Blocks: 50 / 156 (32 %)

Total HSSI RX PCSs: 0 / 6 (0 %)

Total HSSI PMA RX Deserializers: 0 / 6 (0 %)

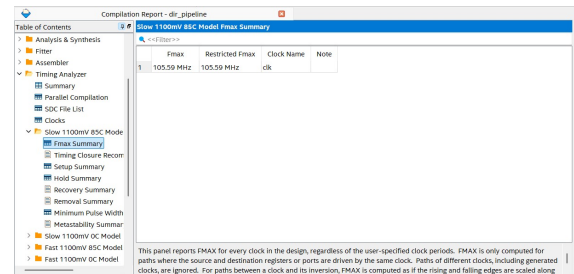
Total HSSI TX PCSs: 0 / 6 (0 %)

Total HSSI PMA TX Serializers: 0 / 6 (0 %)

Total PLLs: 0 / 13 (0 %)

Total DLLs: 0 / 4 (0 %)

(a) Flow Summary



Compilation Report - dir_pipeline

Table of Contents

- Analysis & Synthesis
- Filter
- Assembler
- Timing Analyzer
- Summary
- Parallel Compilation
- SDC File List
- Clocks
- Slow 1100mV 85C Model
- Fmax Summary
- Timing Closure Report
- Setup Summary
- Hold Summary
- Recovery Summary
- Removal Summary
- Minimum Pulse Width
- Metastability Summary
- Slow 1100mV OC Model
- Fast 1100mV OC Model

Slow 1100mV 85C Model Fmax Summary

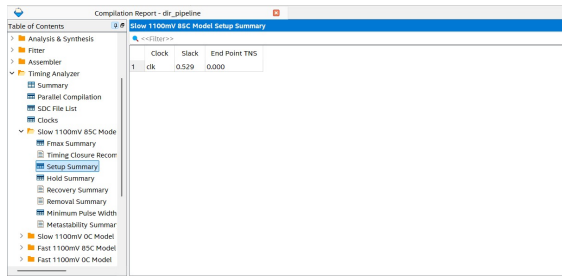
Fmax: 105.59 MHz

Restricted Fmax: 105.59 MHz

Clock Name: clk

Note: This panel reports FMAX for every clock in the design, regardless of the user-specified clock periods. FMAX is only computed for paths where the source and destination registers or ports are driven by the same clock. Paths of different clocks, including generated clocks, are ignored. For paths between a clock and its inversion, FMAX is computed as if the rising and falling edges are scaled along.

(b) Fmax



Compilation Report - dir_pipeline

Table of Contents

- Analysis & Synthesis
- Filter
- Assembler
- Timing Analyzer
- Summary
- Parallel Compilation
- SDC File List
- Clocks
- Slow 1100mV 85C Model
- Fmax Summary
- Timing Closure Report
- Setup Summary
- Hold Summary
- Recovery Summary
- Removal Summary
- Minimum Pulse Width
- Metastability Summary
- Slow 1100mV OC Model
- Fast 1100mV OC Model

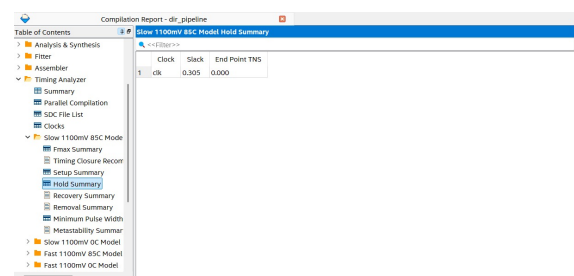
Slow 1100mV 85C Model Setup Summary

Clock: clk

Slack: 0.529

End Point TNS: 0.000

(c) Setup Analysis



Compilation Report - dir_pipeline

Table of Contents

- Analysis & Synthesis
- Filter
- Assembler
- Timing Analyzer
- Summary
- Parallel Compilation
- SDC File List
- Clocks
- Slow 1100mV 85C Model
- Fmax Summary
- Timing Closure Report
- Setup Summary
- Hold Summary
- Recovery Summary
- Removal Summary
- Minimum Pulse Width
- Metastability Summary
- Slow 1100mV OC Model
- Fast 1100mV OC Model

Slow 1100mV 85C Model Hold Summary

Clock: clk

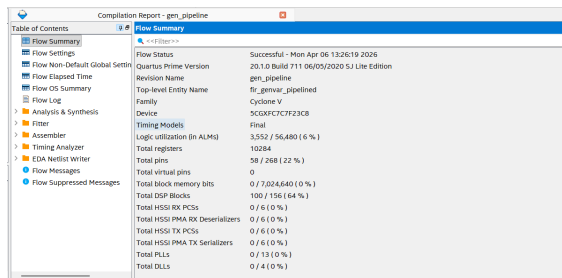
Slack: 0.305

End Point TNS: 0.000

(d) Hold Analysis

Figure 8: Direct Form Pipelined Results

8 Genvar Pipelined



Compilation Report - gen_pipeline

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Settings
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter
- Assembler
- Timing Analyzer
- EDA Netlist Writer
- Flow Messages
- Flow Suppressed Messages

Flow Summary

Flow Status: Successful - Mon Apr 06 13:26:19 2026

Quartus Prime Version: 20.1.0 Build 711 06/05/2020 5J Lite Edition

Revision Name: gen_pipeline

Top-level Entity Name: tr_genvar_pipelined

Family: Cyclone V

Device: 50GXFC7CF23C8

Timing Models: Final

Logic utilization (in ALMs): 3,552 / 56,480 (6 %)

Total registers: 10284

Total pins: 58 / 268 (22 %)

Total virtual pins: 0

Total block memory bits: 0 / 7,024,640 (0 %)

Total DSP Blocks: 100 / 156 (64 %)

Total HSSI RX PCSs: 0 / 6 (0 %)

Total HSSI PMA RX Deserializers: 0 / 6 (0 %)

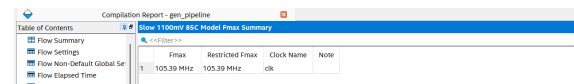
Total HSSI TX PCSs: 0 / 6 (0 %)

Total HSSI PMA TX Serializers: 0 / 6 (0 %)

Total PLLs: 0 / 13 (0 %)

Total DLLs: 0 / 4 (0 %)

(a) Flow Summary



Compilation Report - gen_pipeline

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Settings
- Flow Elapsed Time
- Flow OS Summary

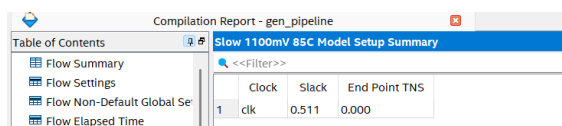
Slow 1100mV 85C Model Fmax Summary

Fmax: 105.39 MHz

Restricted Fmax: 105.39 MHz

Clock Name: clk

(b) Fmax



Compilation Report - gen_pipeline

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Settings
- Flow Elapsed Time

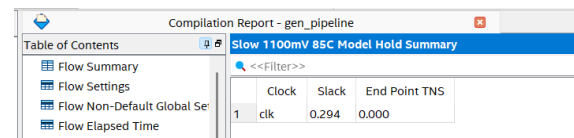
Slow 1100mV 85C Model Setup Summary

Clock: clk

Slack: 0.511

End Point TNS: 0.000

(c) Setup Analysis



Compilation Report - gen_pipeline

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Settings
- Flow Elapsed Time

Slow 1100mV 85C Model Hold Summary

Clock: clk

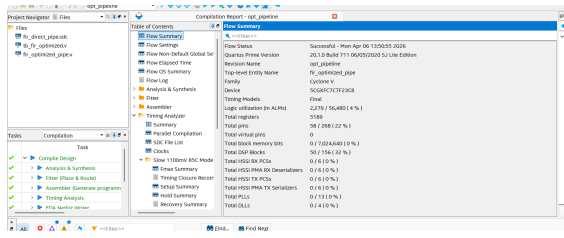
Slack: 0.294

End Point TNS: 0.000

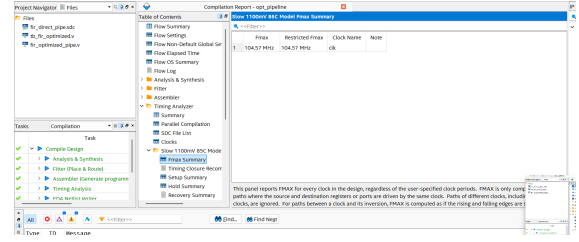
(d) Hold Analysis

Figure 9: Genvar Pipelined Results

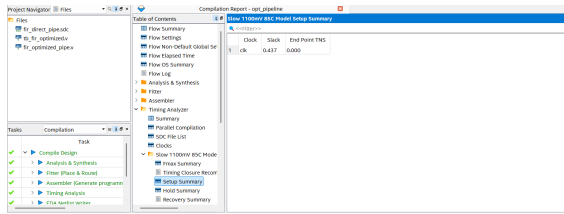
9 Optimized Pipelined



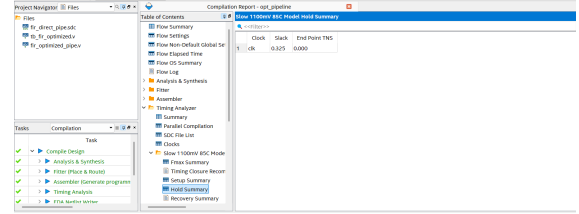
(a) Flow Summary



(b) Fmax



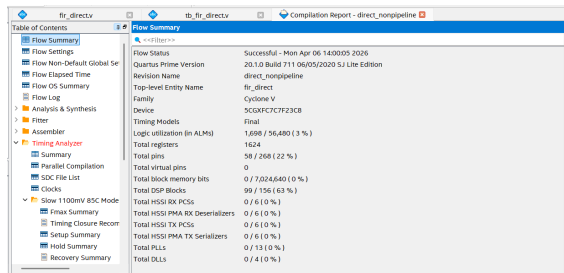
(c) Setup Analysis



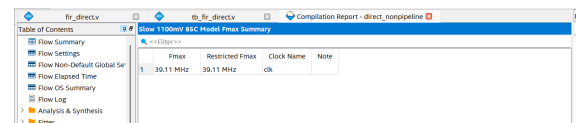
(d) Hold Analysis

Figure 10: Optimized Pipelined Results

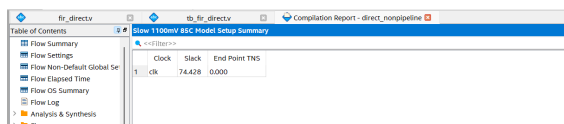
10 Direct Form Non-Pipelined



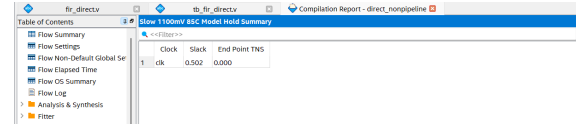
(a) Flow Summary



(b) Fmax



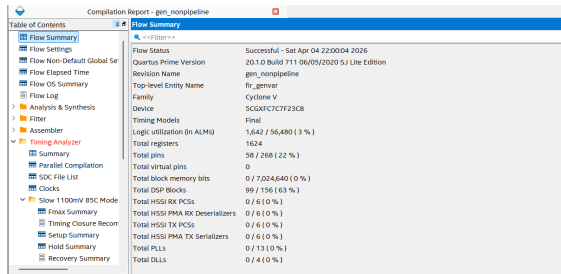
(c) Setup Analysis



(d) Hold Analysis

Figure 11: Direct Form Non-Pipelined Results

11 Genvar Non-Pipelined



Compilation Report - gen_nonpipeline

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Set
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter
- Assembler
- Timing Analyzer
- Summary
- Parallel Compilation
- SDC File List
- Clocks
- Slow 1100mV BSC Mode
- Area Summary
- Timing Closure Recommendation
- Setup Summary
- Hold Summary
- Recovery Summary

Flow Status: Successful - Sat Apr 04 22:00:04 2026

Quartus Prime Version: 20.1.0 Build 711 06/05/2020 S.J. Lite Edition

Revision Name: gen_nonpipeline

Top-level Entity Name: ft_genvar

Family: Cyclone V

Device: 5CGXFC7CF23C8

Timing Models: Final

Logic utilization (in ALM): 1642 / 56,480 (3 %)

Total registers: 1624

Total pins: 58 / 268 (22 %)

Total virtual pins: 0

Total block memory bits: 0 / 7,024,640 (0 %)

Total DSP Blocks: 99 / 156 (63 %)

Total HSSI RX PCSs: 0 / 6 (0 %)

Total HSSI PMA RX Deserializers: 0 / 6 (0 %)

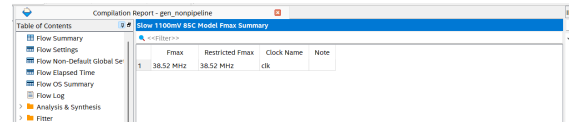
Total HSSI TX PCSs: 0 / 6 (0 %)

Total HSSI PMA TX Serializers: 0 / 6 (0 %)

Total PLLs: 0 / 13 (0 %)

Total DLLs: 0 / 4 (0 %)

(a) Flow Summary



Compilation Report - gen_nonpipeline

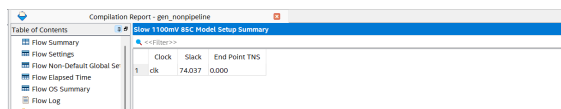
Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Set
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter

Slow 1100mV BSC Model Fmax Summary

Fmax	Restricted Fmax	Clock Name	Note
38.52 MHz	38.52 MHz	clk	

(b) Fmax



Compilation Report - gen_nonpipeline

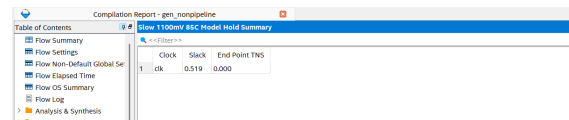
Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Set
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter

Slow 1100mV BSC Model Setup Summary

Clock	Slack	End Point TNS
clk	74.037	0.000

(c) Setup Analysis



Compilation Report - gen_nonpipeline

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Set
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter

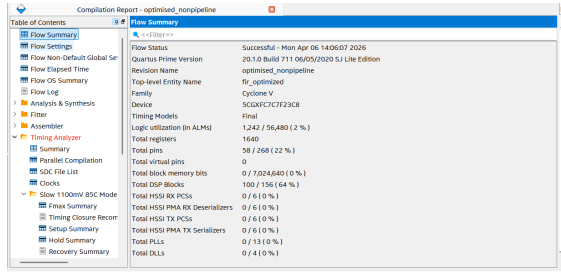
Slow 1100mV BSC Model Hold Summary

Clock	Slack	End Point TNS
clk	0.519	0.000

(d) Hold Analysis

Figure 12: Genvar Non-Pipelined Results

12 Optimized Non-Pipelined



Compilation Report - optimised_nonpipeline

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Set
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter
- Assembler
- Timing Analyzer
- Summary
- Parallel Compilation
- SDC File List
- Clocks
- Slow 1100mV BSC Mode
- Area Summary
- Timing Closure Recommendation
- Setup Summary
- Hold Summary
- Recovery Summary

Flow Status: Successful - Mon Apr 06 14:06:07 2026

Quartus Prime Version: 20.1.0 Build 711 06/05/2020 S.J. Lite Edition

Revision Name: optimised_nonpipeline

Top-level Entity Name: ft_optimized

Family: Cyclone V

Device: 5CGXFC7CF23C8

Timing Models: Final

Logic utilization (in ALM): 1,242 / 56,480 (2 %)

Total registers: 1640

Total pins: 58 / 268 (22 %)

Total virtual pins: 0

Total block memory bits: 0 / 7,024,640 (0 %)

Total DSP Blocks: 100 / 156 (64 %)

Total HSSI RX PCSs: 0 / 6 (0 %)

Total HSSI PMA RX Deserializers: 0 / 6 (0 %)

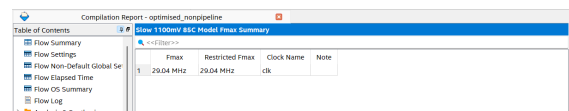
Total HSSI TX PCSs: 0 / 6 (0 %)

Total HSSI PMA TX Serializers: 0 / 6 (0 %)

Total PLLs: 0 / 13 (0 %)

Total DLLs: 0 / 4 (0 %)

(a) Flow Summary



Compilation Report - optimised_nonpipeline

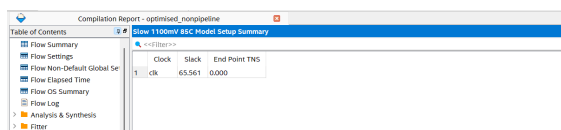
Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Set
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter

Slow 1100mV BSC Model Fmax Summary

Fmax	Restricted Fmax	Clock Name	Note
29.04 MHz	29.04 MHz	clk	

(b) Fmax



Compilation Report - optimised_nonpipeline

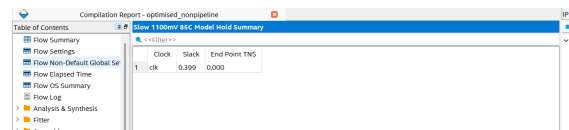
Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Set
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter

Slow 1100mV BSC Model Setup Summary

Clock	Slack	End Point TNS
clk	65.561	0.000

(c) Setup Analysis



Compilation Report - optimised_nonpipeline

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Set
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter

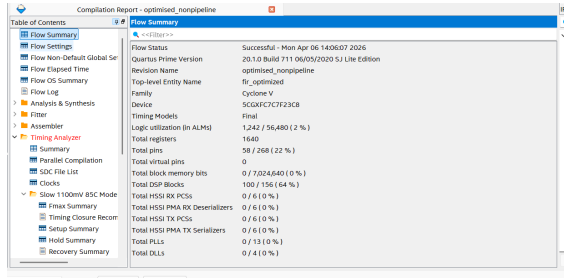
Slow 1100mV BSC Model Hold Summary

Clock	Slack	End Point TNS
clk	0.399	0.000

(d) Hold Analysis

Figure 13: Optimized Non-Pipelined Results

13 Optimized Non-Pipelined



Compilation Report - optimised_nonpipeline

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Flow Summary

Flow Status: Successful - Mon Apr 06 14:06:07 2006

Quartus Prime Version: 20.1.0 Build 711 06/05/2020 S.J. Lite Edition

Revision Name: optimised_nonpipeline

Top-level Entity Name: ft_optimized

Family: Cyclone V

Device: 5C0XFC7CF23C8

Timing Models: Final

Logic utilization (in ALM): 1,242 / 56,480 (2 %)

Total registers: 1640

Total pins: 58 / 268 (22 %)

Total virtual pins: 0

Total block memory bits: 0 / 7,024,640 (0 %)

Total DSP Blocks: 100 / 156 (64 %)

Total HSSI RX PCSs: 0 / 6 (0 %)

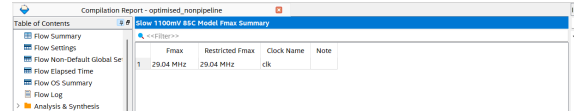
Total HSSI TX PCSs: 0 / 6 (0 %)

Total HSSI PMA RX Deserializers: 0 / 6 (0 %)

Total HSSI PMA TX Serializers: 0 / 6 (0 %)

Total DLLs: 0 / 4 (0 %)

(a) Flow Summary



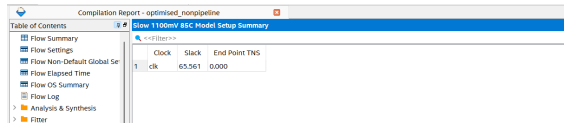
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Slow 1100mV BSC Model Fmax Summary

	Fmax	Restricted Fmax	Clock Name	Note
1	29.04 MHz	29.04 MHz	clk	

(b) Fmax



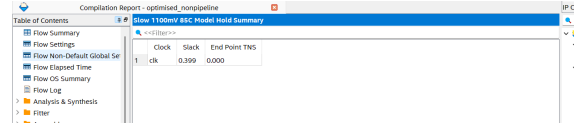
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Slow 1100mV BSC Model Setup Summary

	Clock	Slack	End Point TNS
1	clk	65.561	0.000

(c) Setup Analysis



Compilation Report - optimised_nonpipeline

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Slow 1100mV BSC Model Hold Summary

	Clock	Slack	End Point TNS
1	clk	0.399	0.000

(d) Hold Analysis

Figure 14: Optimized Non-Pipelined Results